KARTHIKA (KATHY) KUMAR

Data Scientist · Data & Cloud Engineer · Forward Deployed Engineer

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PROFILE

Data, AI and software engineer who works where software engineering, data and intelligent systems meet, with three years of professional engineering experience at IBM. On IBM watsonx.data, contributed to enterprise-scale systems spanning vector database integration for generative AI, backend APIs, data pipelines processing up to 1 TB a day, query optimisation and platform security. Alongside this, I build and deploy end-to-end products, from predictive analytics and fraud detection to a live child-health data platform, AI agents and cloud infrastructure. I enjoy taking a complex technical problem, working out what it actually needs to solve, and working across the stack to turn it into a practical, working solution.

CORE SKILLS

Data science: Regression and random forests, LightGBM, spatial and leave-one-group-out cross-validation, calibration, bootstrap stability, small-area shrinkage, geospatial analysis (E2SFCA access), computer vision (YOLOv8, OpenCV)

Data engineering: Python, SQL, PySpark, Databricks (Delta Live Tables, Delta Lake, Unity Catalog), Presto, dbt, DuckDB, Kafka, Apache Iceberg, Airflow, PostgreSQL, Milvus, data-quality testing and lineage

Cloud and software: AWS (Lambda, RDS, Secrets Manager), Docker, Kubernetes, Terraform, GitHub Actions CI/CD, FastAPI, Go REST APIs, C++, gRPC, mutual TLS

Observability and security: Prometheus, OpenTelemetry, Loki, Grafana, audit logging, vulnerability remediation

PROFESSIONAL EXPERIENCE

IBM India Software Labs | **Software Developer, watsonx.data** | Kochi, India — Jan 2022 – Jan 2025

- Built the Milvus connector backend in Python and C++ across SaaS, Cloud Pak for Data and standalone deployments: connection setup, mutual TLS with client certificates, a custom gRPC auth interceptor that refreshed bearer tokens without closing the TLS channel, a pooled connection wrapper for concurrent requests, and test-connection validation and metadata discovery.
- Owned two parts of a banking customer's migration of about 52M vectors from an external store to the platform-native service: the PySpark backfill that read legacy vectors from object storage and ingested them into Milvus in parallel batches, and the shadow-read verification worker that ran production queries against both stores and recorded recall and latency in Prometheus. Cutover followed a 7-day soak at 99.98% recall.
- Cut query time on a 1.2B-row transactions table (about 850 GB) from 24.5 s to 15.8 s (about 35%) using Z-Ordering and then Liquid Clustering on customer_id and transaction_timestamp, reducing files scanned from about 14,200 to about 1,100.
- Developed ETL/ELT pipelines in Databricks, PySpark and SQL over roughly 100 GB to 1 TB per day, using Delta Live Tables across Bronze, Silver and Gold layers, Delta Lake MERGE INTO for incremental loads and Unity Catalog for governance.
- Tuned Presto SQL performance through execution analysis, benchmarking and caching (20% query improvement via open-source contributions), then moved engineering effort to Spark when the team found it more effective for our workloads.
- As security focal for the SaaS and Cloud Pak for Data teams, took on an escalated release blocker: the release was held until security exposure fell from about 30% to under 15%. Closed 42 findings (6 critical, 14 high, 22 medium) over three consecutive releases, implemented audit logging, and replaced a vulnerable log-tracking tool with OpenTelemetry, Loki and Grafana.
- Built Go REST APIs (Bring Your Own JDBC Driver, Test Connection validation across 8 connectors) and automated QA in Python, cutting manual effort by 50% and QA time by 25%.

Junior Engineers | **Coding Instructor and Trainer** | Melbourne — Aug 2025 – present

- Teach three weekly classes of about ten students aged 6 to 12 across Victoria, covering Minecraft cyber security, Python, AI and STEM, and travel to deliver AI classes onsite for older students in Melbourne and Sydney.
- Partner with **Deadly Coders**, which runs coding programs for Aboriginal and Torres Strait Islander learners.
- Set up new classes with the schools I teach at, and trained and onboarded four new instructors on teaching methods, the class portal and attendance.
- Contribute to content planning for upcoming classes.

Monash University, Faculty of IT | **Student Ambassador** | Melbourne — Mar 2026 – present

- Represent Monash at school visits, expos, exhibitions and STEM workshops such as Her Tech Path.
- Allocated student volunteers to orientation placements and coordinated the volunteer team as key ambassador.
- Guided students and answered their questions throughout the Graduate Expo.

PROJECTS

Vkinder: child-health data for parents and practitioners | vkinder.me · Monash Industry Experience capstone 2026

- Data scientist on a team of 6 building a live site that compares child development, services and environmental conditions across 361 Greater Melbourne areas. I built the data pipeline, models and data APIs; teammates built the front end and run production. Finalist, Monash DS&AI Showcase 2026.
- Joined 31 public sources (AEDC, ABS Census, environment, traffic, childcare and health services) onto one map of 361 areas. A weekly GitHub Actions refresh must pass seven quality gates (validity, coverage, key integrity, lineage, reproducibility, freshness, display completeness) or every file rolls back. Suppressed values are kept, never imputed.
- Built a ridge-regression model that predicts each area's expected developmental vulnerability from 10 census measures and flags areas doing worse than expected. The selection rule was written before the results; tested on regions it never saw, its error was 4.1 points against 6.4 with no model (calibration slope 0.99).
- Built the models behind the parent and practitioner tools: a two-step floating catchment (E2SFCA) access score for GPs and childcare, a "Where to live" shortlist with small-area shrinkage and stability labels, and "In 2031" service pressure from official population projections.
- Loaded the validated data into a private, encrypted AWS RDS PostgreSQL database served by FastAPI on AWS Lambda with data contracts. Area-level public data only: no personal data, and no generative AI deciding anything about a child.

Nexora: payments fraud data platform | Python, SQL, Kafka, Apache Iceberg, dbt, Airflow, LightGBM 2026

- Streamed payment events through Kafka into an Apache Iceberg lakehouse with a data contract, quarantine and idempotent reloads, then transformed them in 28 dbt models with data-quality tests, orchestrated with Airflow and tested in GitHub Actions CI.
- Trained a LightGBM model with graph features using time-based validation and 3-seed evaluation: recall 0.97 against 0.45 for SQL rules at the same alert volume (synthetic data), with SHAP explanations in a Streamlit console.

CloudEcho: cloud-native wildlife detection API | YOLOv8, FastAPI, Docker, Kubernetes, Terraform, Azure 2025

- Served a YOLOv8 wildlife-detection model (ONNX Runtime, 320 × 320 input, PyTorch fallback) through FastAPI endpoints for detection and annotated images, with CPU inference moved off the event loop into a thread pool.
- Containerised it with a multi-stage, non-root Docker build and deployed it on Kubernetes on Azure with liveness and readiness probes, CPU and memory limits and 1 to 8 replicas; infrastructure defined in Terraform.
- Load-tested with Locust (70% detect, 30% annotate): throughput rose from 7.58 to 32.32 requests per second between one and four pods, then plateaued at 32.68 on eight, which I reported as the scaling limit.

Voltwise: battery capacity forecasting | Python, scikit-learn, dbt, DuckDB, FastAPI, GitHub Actions 2026

- Forecast lithium-ion cell capacity ten discharge cycles ahead from 636 NASA laboratory discharge cycles across four cells, using rolling-history features and a random forest against persistence and linear-trend baselines.
- Validated leave-one-battery-out over 560 held-out forecasts: MAE 0.024 Ah against 0.044 Ah for persistence (44.7% lower), better in all four folds, with per-cell errors and permutation importance reported.
- Built a dbt + DuckDB metric layer (13 tests), a read-only FastAPI metrics service and a static dashboard with in-browser CSV inference and training-range checks; CI rebuilds and tests the pipeline before every deploy.

EDUCATION

Master of Data Science | Monash University Feb 2025 – Dec 2026

B.Tech (Honours), Computer Science | TKM College of Engineering, Kollam · CGPA 8.61/10 Jul 2018 – May 2022

ACTIVITIES AND ACHIEVEMENTS

- Data and Insight Officer, Monash Data and AI Society (2025)
- Programming Tutor, Faculty of IT, Monash University (2025); Events Officer, Monash Coding Assistive Tech Team (2025)
- Winner, Make-a-thon Hackathon, Cochin University (2019); Winner, IEDC Idea Pitching Contest, Kerala Startup Mission (2021); participant, UNIHACK 2025

CERTIFICATIONS AND WORK RIGHTS

IBM Developer Jumpstart Practitioner (2024); Cybersecurity, Data Privacy and AI, IBM (2024); Security and Privacy by Design (2023); Enterprise Design Thinking: Essentials for AI (2022); Agile Explorer (2022)

Full working rights in Australia (Temporary Graduate visa, subclass 485).

REFERENCES

Ben Tony Joe | **Software Development Manager, IBM India Software Labs** | bentonyjoe191@gmail.com

Ted Li | **Industry Experience Tech Mentor, Monash University** | ted.li@monash.edu